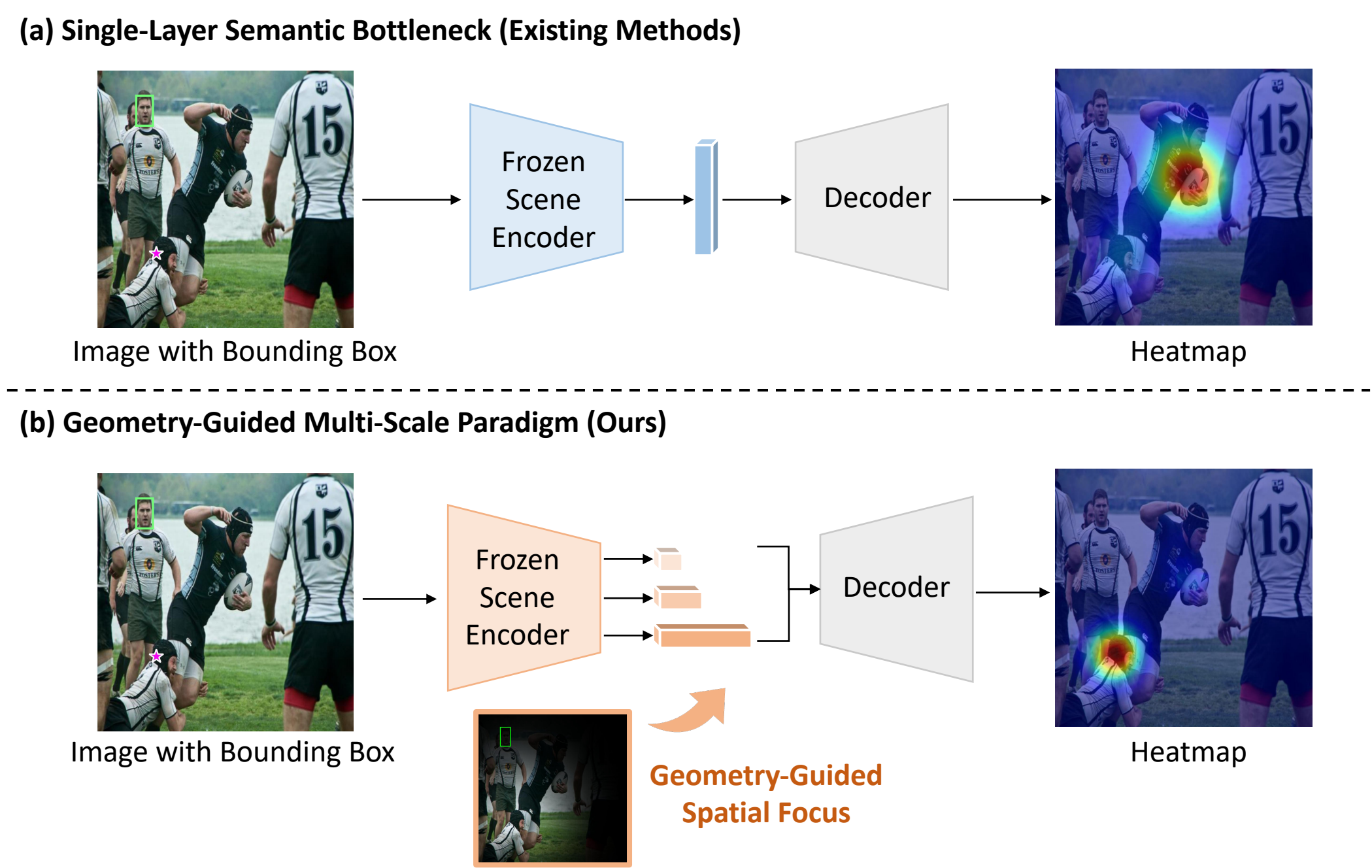

(a) Single-Layer Semantic Bottleneck (Existing Methods)
Frozen Scene Encoder
Decoder
Image with Bounding Box
Heatmap
(b) Geometry-Guided Multi-Scale Paradigm (Ours)
Frozen Scene Encoder
Decoder
Image with Bounding Box
Geometry-Guided Spatial Focus
Heatmap